Koneru Lakshmaiah Education Foundation
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FULL STACK APPLICATION DEVELOPMENT (24SDCS02)

2025-26, Even Semester

Project Problem Statement Submission Form

Sec No-2	**Team No- 9**	**Project Title- Hospital Management System**

1. Abstract

The Hospital Management System is a full stack web application designed to automate and streamline the daily operations of a hospital. Traditional hospital management relies heavily on manual record keeping, which leads to inefficiencies, data redundancy, and difficulty in accessing patient information. This project aims to provide a centralized and user-friendly digital platform to manage hospital activities efficiently.

The proposed system is developed using Spring Boot with Maven for backend processing and Thymeleaf for frontend integration, following a layered architecture. The application enables functionalities such as patient registration, viewing patient records, and maintaining medical data securely using a relational database. The system ensures proper separation of concerns through controller, service, repository, and model layers, improving maintainability and scalability.

The application interacts with a MySQL database using Spring Data JPA and Hibernate for data persistence. The frontend allows users to input patient details through forms and view updated patient records dynamically. This project demonstrates real-world application of full stack development concepts, RESTful architecture, database connectivity, and MVC design pattern. The system can be further extended with features such as doctor management, appointment scheduling, authentication, and billing modules.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Patient Management Module	**2420030239**	**P Hansini**
Backend Logic & Database Integration	**2420030241**	**G Sreevarsha**
Frontend UI (Thymeleaf)	**2420090136**	**V Jasritha**

3. Module Details

- **Patient Management Module:**
Responsible for implementing patient registration, storing patient details, and retrieving patient information from the database.

- **Backend Logic & Database Integration:**
Handles business logic, database connectivity using Spring Data JPA, entity mapping, and service layer implementation.

- **Frontend UI (Thymeleaf):**
Designs user interfaces using HTML and Thymeleaf templates, enabling users to interact with the system through forms and tables.

- **Testing & Documentation:**
Responsible for testing application functionality, validating data flow, and preparing project documentation and reports.

4. Frameworks / Platforms / Software Used

- **Backend:** Spring Boot, Spring Data JPA
- **Frontend:** Thymeleaf, HTML, CSS
- **Build Tool:** Maven
- **Database:** MySQL
- **IDE:** Eclipse IDE
- **Server:** Embedded Apache Tomcat
- **Language:** Java
- **Version Control (optional):** Git

Guide Name and Signature (remarks if any) _____